

EM2020 Probability and Statistics

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References:

- (1) D.C. Montgomery and G.C. Runger (2007). Applied Statistics and Probability for Engineers. 4th Edition (*or any later edition*), John Wiley and Sons, Inc.
- (2) Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 8th edition, (2010), Cengage learning. *Note: e-copy can be downloaded free for some editions.*

Aim and intended learning outcomes (LOs) and LO – PO mapping

Aim: To introduce basic concepts of probability and statistics and their use in applications

Intended Learning Outcomes:

On successful completion of the course, the students should be able to;

- Describe and apply basic concepts of probability and statistics.
- Model and analyze real world problems dealing with uncertainty using the discrete and the continuous random variables.
- Apply concepts of sampling distributions and point estimation for statistical inference and testing hypothesis related to decision making.

EM2020: LO – PO mapping

	Cognitive level	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO1	C1	3	3	0	1	0	0	0	0	0	1	0	1
LO2	C2	3	3	0	1	0	0	0	0	0	1	0	1
LO3	C3	3	3	0	1	0	0	0	0	0	1	0	1
Module		C2,3	C2,3	0	C1,1	0	0	0	0	0	C1,1	0	C1,1

0 – Not covered at all, 1- **Some Exposure**, 2- Explicitly taught, 3- **Explicitly taught and assessed**, 4-Assessed only (assumed learnt earlier)

C1 – Remembering and Understanding, C2 – Applying, C3 – Analyzing and Evaluating, C4 – Creating,

PO1 Apply knowledge of mathematics, basic sciences and engineering fundamentals to solve complex engineering problems.

PO2 Identify, formulate, research literature, and analyze complex engineering problems to arrive at valid conclusions.

PO4 Conduct investigations of complex engineering problems using research based knowledge and research methods (including experimental design, analysis & interpretation of data and synthesis of information).

PO10 Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO12 Be prepared for independent and lifelong learning in the broad context of technological change.

80% Course Participation

The time allocation for this course is,

24 Lecture hours (L), 4 Tutorial hours (T) and 4 Assignment hours (A).

$$\text{Course Participation (Attendance)} = \frac{L_s + T_s + 0.5A_s}{30} \times 100 \geq 80\%$$

Where, L_s , T_s and A_s are the students participation in the above components.

Note that each and every component is considered equally important.

Assessment criteria

Assessment	Percentage mark
Tutorials and assignments	10
Mid semester examination	30
End semester examination	60

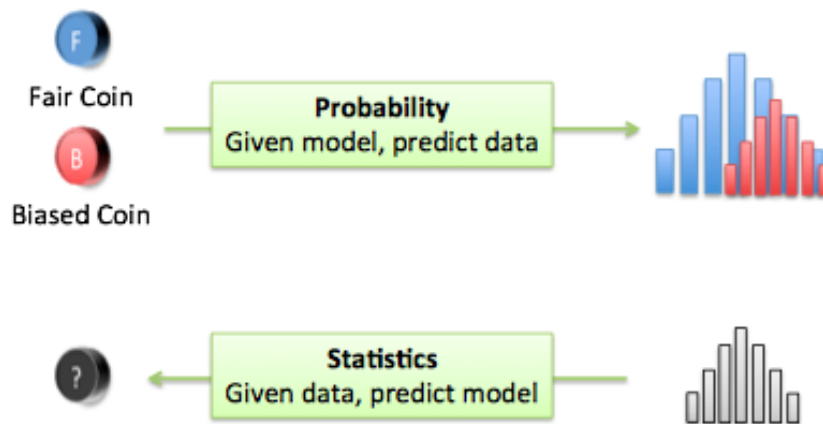


Figure I: The roles of probability and statistics in applications.

Some quotes related to probability and statistics

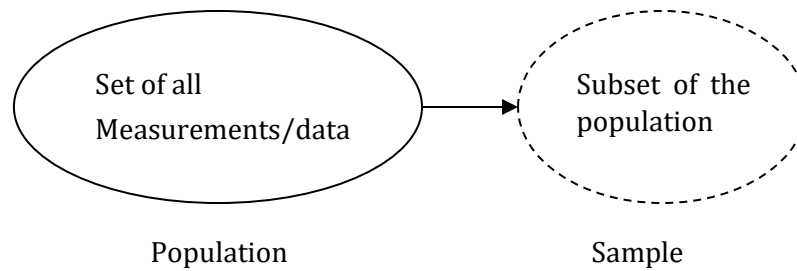
“Statistics is a method of decision-making in the face of uncertainty on the basis of numerical data and calculated risks.” — **Prof. YA LU CHOU**

“It is a part of probability that many improbable things will happen.”

—**Aristotle**

1 Sampling distributions

In practical problems, we often need to summarize and use the information contained in large sets of measurements (data sets). Most such cases involve sampling. In sampling, a specified number of items, i.e. a sample, is drawn from a much larger body of data called the population. This sample is used to make inferences about the population.



Population:

A population is the set representing all observations of interest to the sample collector.

Sample:

A sample is a subset of measurements selected from the population of interest.

Sampling Methods:

- e.g. Simple random sampling,
- Stratified random sampling,
- Cluster sampling

Simple random sampling:

In simple random sampling each member of the population has an equal chance of being selected.

Random sample:

The random variables $X_1, X_2, \dots, X_n$ are a random sample of size n if,
the X_i 's are independent random variables, and
every X_i has the same probability distribution.

Statistic:

A statistic is a function of the observations in a random sample.

e.g. Sample mean: $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$
Sample variance: $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$

Note: A statistic is a random variable; therefore, it has a probability distribution.

Sampling distribution:

The probability distribution of a statistic is called a sampling distribution.

1.1 Sampling distribution of sample mean ($\bar{X}$)

The sample mean $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$

The probability distribution of $\bar{X}$ is called sampling distribution of the sample mean.

Sampling distribution of $\bar{X}$ when a random sample taken from a normal population

Suppose a random sample of size n is taken from a normal population with mean μ and variance σ^2 . Each observation in this sample say $X_1, X_2, \dots, X_n$ is a normally and independently distributed random variable with mean μ and variance σ^2 . Then the sample mean, $\bar{X}$, is normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.

Note that:

$$E[\bar{X}] = \mu_{\bar{X}} = \frac{\mu + \mu + \dots + \mu}{n} = \mu$$
$$V[\bar{X}] = \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n^2} + \frac{\sigma^2}{n^2} + \dots + \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Therefore, we get $\rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

It has been shown that for a random sample taken from a population with unknown probability distribution also the sampling distribution of $\bar{X}$ is still approximately normal with mean μ and variance $\frac{\sigma^2}{n}$, if the sample size n is large. This result is given by the central limit theorem.

Ex 1.1 Consider a large population which consists of equal numbers of digits 1, 2 and 3.

- List all possible samples of size 2.
- Find their sample means.
- Find the sampling distribution of the sample mean.
- Find the mean and variance of the sample mean, $\bar{X}$.
- Find the mean and variance of the population and compare them with those of $\bar{X}$.

Ex 1.2 An electronics company manufactures resistors that have a mean resistance of 100Ω and a standard deviation of 10Ω . The distribution of resistance is normal.

- Find the probability that a random sample of size $n = 25$ resistors will have an average resistance less than 95Ω .
- Plot the probability distributions of
 - The resistance and
 - the sample mean of the resistors.

1.2 Central limit theorem

If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2 , and if $\bar{X}$ is the sample mean, the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad \text{as } n \rightarrow \infty, \quad \text{is the standard normal distribution.}$$

For practical purposes, $n > 30$ is considered sufficient to apply CLT and assume that of $\bar{X}$ is approximately normal, i.e., $\bar{X} \sim N(\mu, \sigma^2/n)$.

Ex 1.3 In Ex 1.2, assume that the distribution of resistance is uniform. Suppose a random sample of $n = 36$ resistors is taken,

- (a) What is the probability that the average resistance of the sample is less than 98Ω ?
- (b) Sketch the probability distributions of
 - (i) the resistance and
 - (ii) the sample mean of the resistors.

1.3 The sampling distribution of the sample variance (S^2)

Sample variance: $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2 , so that $Z_i = \frac{X_i - \mu}{\sigma} \sim N(0,1)$.

- $Z_i^2 = \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi_1^2$
- $\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi_n^2$
- $\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi_{n-1}^2$

$$\boxed{\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2}$$

Ex 1.4 Compressive strength of some material is normally distributed with $\mu = 250\text{psi}$ and $\sigma = 10\text{psi}$. Find the value d such that the probability of the sample variance S^2 of a random sample of $n = 26$ specimens falling below d is 0.1.

2 Inferences on mean and variance

2.1 Point Estimation

Probabilistic models for populations usually contain unknown parameters, which are called population parameters. Statistics may be used to estimate these unknown parameters. That is, to obtain a reasonable value (or guess) of the true values.

e.g. Assume that the height of a university student is $\sim N(\mu, \sigma^2)$. Here μ, σ^2 are unknown population parameters. The sample mean is an estimate for the population mean. Sample variance is the estimate of the population variance.

Point estimator

A point estimator is a statistic that specifies how to use the sample data to estimate an unknown parameter of the population. A point estimator is a random variable.

If X is a random variable with probability distribution $f(x)$ with population parameter θ and if $X_1, X_2, \dots, X_n$ is a random sample of size n from X , the statistic $\hat{\theta} = h(X_1, X_2, \dots, X_n)$ is called a point estimator of θ .

After a sample is selected, $\hat{\theta}$ takes on a particular numerical value $\hat{\theta}$ that is called the point estimate of θ .

Ex 2.1 Suppose the random variable X is normally distributed with an unknown mean μ . A sample of $x_1 = 25, x_2 = 30, x_3 = 29$ and $x_4 = 31$ has been selected.

- (a) Determine a suitable point estimator to the unknown population mean μ assuming a random sample of size 4.
- (b) Determine the point estimate of μ .

Some point estimators for population parameters

- For mean μ the estimator is $\bar{X}$ and the estimate is $\bar{x}$, the sample mean
- For variance σ^2 the estimator is S^2 , the estimate is s^2 , the sample variance
- For the difference in means of two populations $\mu_1 - \mu_2$, the estimator is $\bar{X}_1 - \bar{X}_2$, the estimate is $\bar{x}_1 - \bar{x}_2$, the difference between the sample means.

Unbiased estimators

The point estimator $\hat{\theta}$ is an unbiased estimator for the population parameter θ if

$$E[\hat{\theta}] = \theta.$$

If the estimator is not unbiased, then the difference $E[\hat{\theta}] - \theta$ is called the bias of the estimator $\hat{\theta}$.

Normally, it is desirable that all possible values of $\hat{\theta}$ to be very close to θ . Therefore, we prefer smaller variance for $\hat{\theta}$.

Minimum Variance Unbiased Estimator (MVUE)

If we consider all unbiased estimators of θ the one with the smallest variance is called the minimum variance unbiased estimator (MVUE)

e.g. If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a normal distribution, the sample mean $\bar{X}$ is the MVUE for the mean.

Ex 2.2 Suppose $X_1, X_2, \dots, X_n$ is a random sample from some normal population with $E[X_i] = \mu$ and $V[X_i] = \sigma^2$ for $i = 1, 2, \dots, 5$. Which estimator for μ is preferred from the followings?

- (i) $\hat{\theta}_1 = X_1$
- (ii) $\hat{\theta}_2 = \frac{1}{2}(X_1 + X_2)$
- (iii) $\hat{\theta}_3 = \frac{1}{5}(X_1 + X_2 + X_3 + X_4 + X_5)$
- (iv) $\hat{\theta}_4 = 2X_1 - X_2$

Ex 2.3 Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2 .

- (a) Show that $\bar{X}^2$ is a biased estimator for μ^2 .
- (b) Find the amount of bias in this estimator.
- (c) What happens to the bias as the sample size n increases?

Mean square error of an estimator

The mean square error of an estimator $\hat{\theta}$ is defined as,

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

This may be rewritten as,

$$MSE(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2] + (\theta - E[\hat{\theta}])^2$$

$$MSE(\hat{\theta}) = V[\hat{\theta}] + bias^2$$

When comparing two estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of the parameter θ , where $MSE(\hat{\theta}_1)$ and $MSE(\hat{\theta}_2)$ are the mean square errors of them, the relative efficiency of $\hat{\theta}_2$ to $\hat{\theta}_1$ is defined as $\frac{MSE(\hat{\theta}_1)}{MSE(\hat{\theta}_2)}$. If this relative efficiency is less than one, $\hat{\theta}_1$ is a more efficient estimator of θ than is $\hat{\theta}_2$, in the sense that it has smaller mean square error.

2.2 Confidence Intervals (Interval estimates)

Bounds that represent an interval of plausible values for a parameter is an interval estimate. An interval estimate for a population parameter is called a “confidence interval”.

In many cases such intervals can be determined from the same data that provide the point estimate.

E.g. Determining the mean height of students (μ) in the Peradeniya University using a sample.

2.2.1 Development of a confidence interval

A confidence interval estimate for a population parameter θ is an interval,

$$l \leq \theta \leq u$$

where l is a lower limit and u is an upper limit computed from the sample data.

Since l and u depend on sample data, these end points are values of random variables L and U . We determine values l and u of L and U respectively such that,

$$P(L \leq \theta \leq U) = (1 - \alpha).$$

The value $100(1 - \alpha)\%$ is called the confidence level.

Confidence bounds

For a population parameter θ , a confidence bound of the form,

$l \leq \theta$ is called a lower confidence bound, and a bound of the form,

$\theta \leq u$ is called an upper confidence bound, and that of the form.

The values of l and u in the confidence bounds are determined such that

$$P(L \leq \theta) = (1 - \alpha) \quad \text{and} \quad P(\theta \leq U) = (1 - \alpha).$$

Common confidence levels used in practice are 90%, 95%, and 99% that result from α values 0.1, 0.05 and 0.01 respectively.

2.2.2 Confidence Interval on the Mean of a normal distribution when variance is known

Suppose we have a normal population with unknown mean μ and known variance σ^2 . Let $X_1, X_2, \dots, X_n$ be a random sample from the population. An interval estimate for μ with a given level of confidence can be given as follows.

If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2 , a $100(1 - \alpha)\%$ Confidence Interval (CI) on μ is given by,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the upper $100\alpha/2$ percentage point of the standard normal distribution.

Similarly, a $100(1 - \alpha)\%$ upper confidence bound (CB) for μ is,

$$\mu \leq \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

and a $100(1 - \alpha)\%$ lower confidence bound for μ is,

$$\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}} \leq \mu.$$

Ex 2.4 The Charpy V-notch (CVN) technique is used to determine whether or not a material experience a ductile-to-brittle transition with decreasing temperature. Ten measurements of impact energy (J) on specimens of A238 steel cut at 60 °C are: 64.1, 64.7, 64.3, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2 and 64.3. Assuming that the impact energy is normally distributed with $\sigma = 1$ J find a 95% confidence interval for the mean impact energy.

Interpretation of a confidence interval

Since L and U (defining end points) are random variables, confidence interval is a random interval. In practice we obtain one random sample and calculate one confidence interval. The appropriate way to interpret this is, the observed interval $[l, u]$ encloses the true value of μ with confidence $100(1 - \alpha)\%$. This means, we do not know if the statement is true for this specific sample, but the method used to obtain the interval $[l, u]$ yields correct statements $100(1 - \alpha)\%$ of the time.

Choice of sample size

If $\bar{x}$ is used as an estimate of μ we can be $100(1 - \alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is,

$$n = \left(\frac{Z_{\alpha/2} \sigma}{E} \right)^2.$$

The length of resulting Confidence Interval is $2E$.

Ex 2.5 In the previous example on CVN test, determine how many specimens must be tested to ensure that the 95% CI on the mean impact energy for A238 steel cut at 600C has a length of at most 1.0 J?

2.2.3 Confidence intervals on the mean of a normal population, variance unknown

In this case too, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$. However, if the unknown σ is replaced by S , the random variable $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ will no longer be normally distributed, especially when n is small.

***t*- Distribution**

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with unknown mean μ and unknown variance σ^2 . The random variable,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t - distribution with $n - 1$ degrees of freedom.

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown σ , a $100(1 - \alpha)\%$ confidence interval on μ is given by,

$$\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2, n-1}$ is the upper $100\alpha/2$ percentage point of the t - distribution with $n - 1$ degrees of freedom.

This is useful especially for small samples taken from normal populations.

Ex 2.6 Answer the followings.

- Find the t -value with 5 d.o.f and an α value of 0.1
- Find the t -value with 23 d.o.f and an α value of 0.025
- What is the distribution of T when $\rightarrow \infty$?

Ex 2.7 An article in the journal of Materials Engineering describes the results of tensile adhesion tests on 22 U-700 alloy specimens. The load at specimen failure (in MPa) is as follows.

19.8 10.1 14.9 7.5 15.4 15.4
 15.4 18.5 7.9 12.7 11.9 11.4
 11.4 14.1 17.6 16.7 15.8
 19.5 8.8 13.6 11.9 11.4

Assuming that the population is normally distributed,

(a) find a 95% CI on the mean load at specimen failure.

(b) find a 95% lower confidence bound for the mean load at specimen failure.

2.2.4 Large - Sample Confidence interval for mean

Case 1: when σ^2 is known

By central limit theorem the sample mean $\bar{X}$ from any population has a normal distribution when n is large. Therefore, a $100(1 - \alpha)\%$ CI on μ is,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Case 2: when σ^2 is unknown

When n is large, replacing σ by the sample standard deviation S in $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ has little effect on its distribution. Therefore, $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is approximately a standard normal distribution.

Consequently, a $100(1 - \alpha)\%$ CI on μ is,

$$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

Central limit theorem holds for $n \geq 30$, but to use the above result, $n \geq 40$ is recommended since replacing σ by S results in additional variability.

Table 2.1 Summary: $100(1 - \alpha)\%$ Confidence intervals on mean

Distribution of Population, X Mean: μ Variance: σ^2	σ^2	n	Sampling Distribution of $\bar{X}$	Confidence intervals on μ
Normal	Known	Small/ Large	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
	Unknown	Small/ Large	$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$	$\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
Any distribution	Known	Large ($n \geq 30$)	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
	Unknown	Large ($n \geq 40$)	$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0,1)$ Approximately	$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}$

Table 2.2 Summary: $100(1 - \alpha)\%$ Confidence bounds on mean

Distribution of Population, X Mean: μ Variance: σ^2	σ^2	n	Sampling Distribution of $\bar{X}$	Confidence bounds on μ	
				Lower CB	Upper CB
Normal	Known	Small/ Large	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}} \leq \mu$	$\mu \leq \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}$
	Unknown	Small/ Large	$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$	$\bar{x} - t_\alpha \frac{s}{\sqrt{n}} \leq \mu$	$\mu \leq \bar{x} + t_\alpha \frac{s}{\sqrt{n}}$
Any distribution	Known	Large ($n \geq 30$)	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}} \leq \mu$	$\mu \leq \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}$
	Unknown	Large ($n \geq 40$)	$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0,1)$ Approximately	$\bar{x} - z_\alpha \frac{s}{\sqrt{n}} \leq \mu$	$\mu \leq \bar{x} + z_\alpha \frac{s}{\sqrt{n}}$

2.2.5 Confidence interval for the population variance σ^2 when the population is normally distributed

Let $X_1, X_2, \dots, X_n$ be a random sample taken from a normal distribution.

Recall the sampling distribution of S^2 : $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

$$\begin{aligned}
 P(L \leq \sigma^2 \leq U) &= 1 - \alpha \\
 P\left(\chi_{1-\alpha/2, n-1}^2 \leq \left(\frac{n-1}{\sigma^2}\right) S^2 \leq \chi_{\alpha/2, n-1}^2\right) &= 1 - \alpha \\
 P\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}\right) &= 1 - \alpha
 \end{aligned}$$

The $100(1 - \alpha)\%$ confidence interval for population variance is,

$$\boxed{\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}}$$

Ex 2.8 An automatic filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles gave $s^2 = 0.0153$ ounces² for fill volume.

- Find a 95% CI for the variance of the fill volume.
- Find a 95% upper confidence bound for the standard deviation of the fill volume.

3 Introduction to Hypothesis Testing

hypothesis (noun)

An idea or explanation of something that is based on a few known facts but that has not yet been proved to be true or correct. (*from the Oxford Advanced Learner's Dictionary*)

In hypothesis testing, we use sample data to choose between two competing hypotheses, called the null hypothesis and the alternative hypothesis. We assume that the null hypothesis is true until the alternative is shown beyond "chance".

e.g. Think of a jury trial where,
Null hypothesis (H_0) : Innocent
Alternative hypothesis (H_1): Guilty
You assume innocence until shown guilty beyond a reasonable doubt.

Null Hypothesis - the status quo - initially assumed true
Alternative Hypothesis - the researcher's proposal (what you hope to show)

In hypothesis testing we try to reject the null hypothesis in favor of the alternative only with convincing or significant evidence.

The two possible decisions of the test are, "Reject H_0 " or "Fail to reject H_0 ".

Interpretation:

In the earlier example, "failure to reject H_0 " does not imply innocence. It is just the evidence was insufficient to convict.

On the other hand, when we "reject H_0 ", it does not imply H_1 is necessarily true. It only says we have significant evidence to reject the null.

Forms of Hypotheses:

Null Hypothesis:

H_0 : population parameter = some hypothesized value

Alternative Hypothesis:

H_1 : population parameter $\neq$ the hypothesized value (two – tailed) or,

H_1 : population parameter $>$ the hypothesized value (right tailed) or,

H_1 : population parameter $<$ the hypothesized value (left tailed)

$H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$, or $H_0: \theta > \theta_0$ or $H_0: \theta < \theta_0$

Ex 3.1 The mean height of Sri Lankan males (μ) is reported as 165 cm. Write the null and the alternative hypothesis for the following tests.

(a) Test whether the mean height is different from 165 cm.

(b) Test whether the mean height is greater than 165 cm.

(c) Test whether the mean height is less than 165 cm.

Answer:

(a) $H_0: \mu = 165$ vs $H_1: \mu \neq 165$

(b) $H_0: \mu = 165$ vs $H_1: \mu > 165$

(c) $H_0: \mu = 165$ vs $H_1: \mu < 165$

The null hypothesis implicitly maintains values not mentioned in the alternative hypothesis. For example if $H_1: \mu > 12$ then the null hypothesis is written as $H_0: \mu = 12$, but it could also be written as $H_0: \mu \leq 12$.

Ex 3.2 Each of the following paragraphs calls for a statistical test about a population mean μ . State the null hypothesis H_0 and the alternative hypothesis H_1 in each case.

- A company claims that there are 20 oz in their soda bottles. You (a consumer) will sue if there is less than 20 oz. on average per bottle. What hypotheses should you test?
- The diameter of a spindle in a small motor is supposed to be 5 mm. If the spindle is either too small or too large, the motor will not work properly. The manufacturer measures the diameter in a sample of motors to determine whether the mean diameter has moved away from the target.
- Census Bureau data show that the mean household income in the area served by a shopping mall is \$42,500 per year. A market research firm questions shoppers at the mall. The researchers suspect that the mean household income of mall shoppers is higher than that of the general population.
- The examinations in a large accounting class are scaled after grading so that the mean score is 50. The professor thinks that one teaching assistant is a poor teacher and suspects that his students have a lower mean score than the class as a whole. The TA's students this semester can be considered a sample from the population of all students in the course, so the professor compares their mean score with 50.

3.2 Errors in hypothesis testing

Hypotheses are “rejected” or “fail to reject” based on a confidence level which is less than 100%. Therefore, there is a possibility for error in the decision. There are two types of possible errors.

Type I Error

Rejecting the null hypothesis when in fact, the null hypothesis is true.

e.g. Innocent man is pronounced guilty.

Type II Error

Not rejecting the null hypothesis when in fact, the alternative is true.

e.g. Guilty man is pronounced innocent.

Type I and Type II errors have different consequences. Type I error is usually considered more serious.

Table 3.1 Possible situations in testing a statistical hypothesis

Decision	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

Related quantities:

- Probability of a Type I Error = $P(\text{Type I Error}) = \alpha$
- Probability of a Type II Error = $P(\text{Type II Error}) = \beta$
- Significance level α is the probability of a Type I error.
(One has to set this in advance of the test)
- Power of the test = $1 - \beta$

Power refers to the probability of rejecting the null hypothesis when the alternative is true. Ideally, you want power to be as large (close to 1) as possible. There are two ways to increase the power:

1. Increase sample size, n
2. Increase the significance level, α .
(Note that you still don't want to set α too high).

An Example on Errors and their Consequences:

You want to try a new expensive cancer treatment for your patients. The current treatment has a remission rate of 40 percent. You want to see if the new treatment is more effective. That is,

$$H_0: p = 0.4 \quad \text{vs} \quad H_1: p > 0.4$$

where p is the population proportion of cancer patients who enter remission after undergoing the new treatment.

A Type I error in this context would mean that you concluded that the new treatment is more effective than the current treatment, when in fact, it isn't more effective. As a consequence, you may assign patients to the new treatment, which is expensive, when it is not any better than the current treatment.

A Type II error in this context would mean that you concluded that the new treatment is not more effective than the current treatment, when in fact, it is more effective. As a consequence, you may not switch patients to the new treatment which would give them a better chance at remission than the current treatment.

Ex 3.3 We want to explore the mean number of eggs laid by female fish of a certain species because we are concerned about the survival of the species. If the average number of eggs laid is less than 25, the species will be considered endangered and precautions to help it survive may be implemented. What hypotheses should you test? (Define the population parameter of interest.) Interpret what Type I and Type II Errors mean in context of this example and give a consequence associated with each.

3.3 Steps in Hypothesis Testing

1. State the null and alternative hypothesis.
2. Choose a fixed significance level α .
3. Choose an appropriate test statistic and establish the critical region based on α .
4. From the computed test statistic, reject H_0 if the test statistic is in the critical region. Otherwise do not reject.
5. Draw scientific or engineering conclusions.

3.4 Tests concerning a single mean

Suppose it is required to test whether the mean of a population is different from the hypothesized value μ_0 .

Case 1: A random sample $X_1, X_2, \dots, X_n$ is taken from a population that is normally distributed with mean μ and variance σ^2 .

(a) Population variance known

Null and the alternative hypothesis are:

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0.$$

Let us take the significance level for the test as α .

Under H_0 , we have

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

Critical region:

$$\bar{x} \leq \mu_0 - z_{\alpha/2} \sigma / \sqrt{n} \quad \text{or} \quad \bar{x} \geq \mu_0 + z_{\alpha/2} \sigma / \sqrt{n}$$

Or equivalently,

$$z \leq -z_{\alpha/2} \quad \text{or} \quad z \geq z_{\alpha/2}$$

Make a decision:

- If the computed test statistic $z_c = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ is in the critical region - Reject H_0
- Otherwise, do not reject H_0 .

Alternatively,

- Reject H_0 if $\bar{x} \leq \mu_0 - z_{\alpha/2} \sigma / \sqrt{n}$ or $\bar{x} \geq \mu_0 + z_{\alpha/2} \sigma / \sqrt{n}$.
- Otherwise do not reject H_0 .

(b) Population variance unknown

The procedure can be followed with t - distribution.

The sampling distribution is, $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$

The computed test statistic is, $t_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Case 2: Large samples.

A random sample $X_1, X_2, \dots, X_n$ is taken from a population with mean μ and variance σ^2 .

(a) Population variance known

The sampling distribution is, $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$

The computed test statistic is, $z_c = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

(b) Population variance unknown

The sampling distribution is, $Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim N(0, 1)$

The computed test statistic is, $z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

***P*-value**

The *P*-value is the smallest level of significance that would lead to rejection of the null hypothesis with the given data.

- Ex 3.4** An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed. It is claimed that the light bulbs have a mean lifetime of 800 hours and a standard deviation of 40 hours. Test the hypothesis that the mean lifetime is different from 800 hours if a random sample of 30 bulbs has an average life of 788 hours. Use a 0.05 level of significance.
- Ex 3.5** A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.
- Ex 3.6** The Edison Electric Institute published figures on the annual number of kilowatt hours expended by various home appliances. It is claimed that a vacuum cleaner expends an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners expend an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners expend, on the average, less than 46 kilowatt hours annually? Assume that the population of kilowatt hours expended by vacuum cleaners to be normal.
- Ex 3.7** What is the smallest value of α that you can reject H_0 in Ex 3.5?
- Ex 3.8** Calculate the *P*-values for the tests in Ex 3.4 and Ex 3.6?

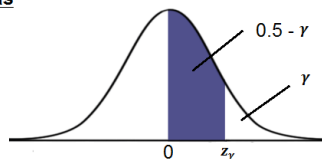
Table 3.2 *P*-values for different tests

Test	Hypothesis	P-value
Two tailed	$H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$	$2P(Z > z_c)$
Upper tailed (Right tailed)	$H_0: \mu = \mu_0$ $H_1: \mu > \mu_0$	$P(Z > z_c)$
Lower tailed (Left tailed)	$H_0: \mu = \mu_0$ $H_1: \mu < \mu_0$	$P(Z < z_c)$ $P(Z > z_c)$

Appendix A: Statistical tables

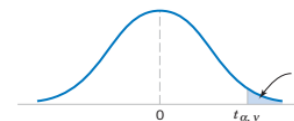
A.1 Table: Standard normal curve areas

Normal curve areas



z_y	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

A.2 Table: Percentage points of the t - distribution



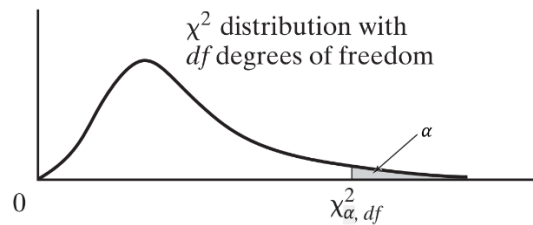
Percentage Points $t_{\alpha, v}$ of the t Distribution

α	.40	.25	.10	.05	.025	.01	.005	.0025	.001	.0005
1	.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	.289	.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	.277	.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	.271	.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	.267	.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	.265	.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	.263	.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	.262	.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	.261	.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	.260	.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	.260	.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	.259	.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	.259	.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	.258	.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	.258	.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	.258	.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	.257	.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	.257	.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	.257	.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	.257	.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	.257	.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	.256	.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	.256	.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	.256	.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	.256	.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	.256	.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	.256	.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	.256	.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	.256	.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	.256	.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	.255	.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	.254	.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	.254	.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	.253	.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

v = degrees of freedom.

A.3 Table: Chi – squared distribution

Values of $\chi^2_{\alpha, df}$ such that $P(\chi^2_{df} > \chi^2_{\alpha, df}) = \alpha$



Degrees of Freedom (df)	Area to the Right of Critical Value (α)									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169